TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of the Strength of a Supplied Hydrogen Peroxide Solution

Peroxide turns reductant and hands its oxygen to permanganate — a titration that reports strength twice over: in grams per litre, and in the 'volumes' of oxygen printed on every commercial bottle.

CHE-1206 · CHEMISTRY-II SESSIONAL · EXPERIMENT 9 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to find the strength of a supplied hydrogen peroxide solution by titration with standard permanganate. Hydrogen peroxide is usually an oxidising agent, but against the stronger oxidant $KMnO_4$ it plays the opposite role — it is oxidised, and its 'available oxygen' leaves as gas:

$$2\ MnO_4^- + 5\ H_2O_2 + 6\ H^+ \rightarrow 2\ Mn^{2+} + 5\ O_2 + 8\ H_2O \quad \text{(2 mol permanganate : 5 mol peroxide)}$$

The reaction is fast in the cold (Mn^{2+} autocatalysis again helps it along), needs no indicator — permanganate is self-indicating as before — and must **not** be heated: peroxide decomposes on warming, so the heating trick of Experiment 7 would destroy the analyte it is meant to measure. Because the supplied solution is too strong to titrate directly (the titre would run off the burette), it is first diluted by a known factor and the aliquot titrated.

Two ways of stating the answer. Strength in **g L^{-1}** is the chemist's figure. The **volume strength** is the trade's: the number of litres of oxygen (at STP) that one litre of the solution releases on decomposition, from $2\ H_2O_2 \rightarrow 2\ H_2O + O_2$. The two are locked together — a '10-volume' solution is about 30 g L^{-1} — and this experiment reports both.

2. Apparatus and solutions

- Standard $KMnO_4$ solution, 0.0200 M (standardised in Experiment 7), in the burette.
- The supplied H_2O_2 solution; pipettes 10 mL and 25 mL; 250 mL volumetric flask.
- Dilute sulphuric acid (about 2 M); conical flask, white tile.

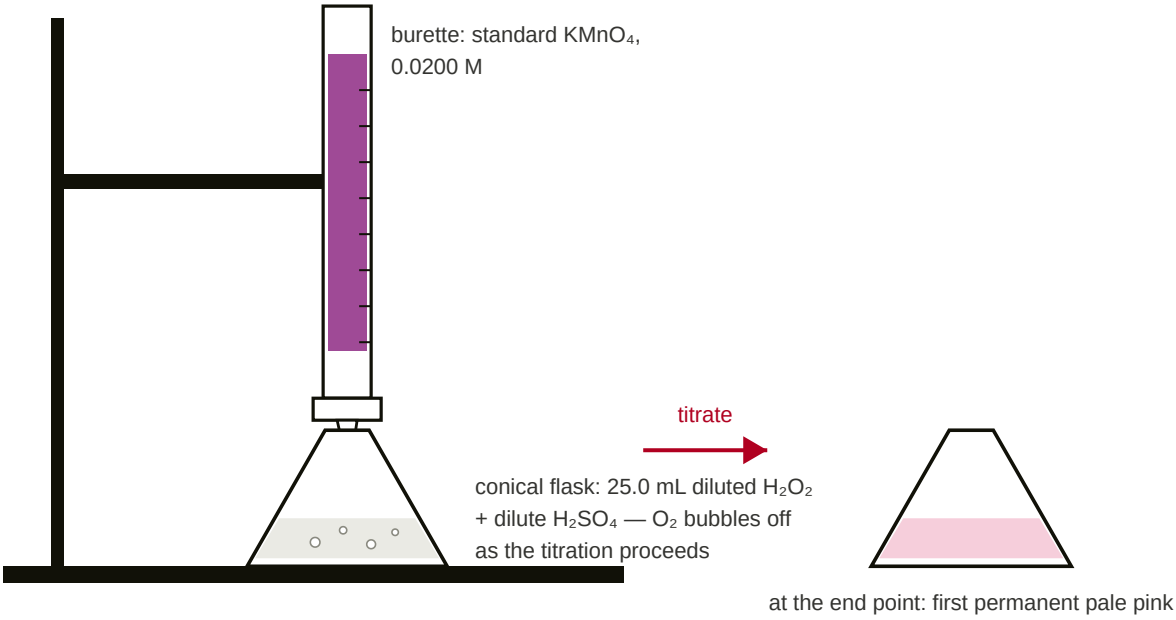


Figure 1 — Titrating diluted peroxide with standard permanganate. Each portion of MnO_4^- is discharged by peroxide and the oxygen escapes as bubbles; the first drop with no peroxide left to meet it stays pink.

3. Procedure

1. Pipette 10.0 mL of the supplied H_2O_2 solution into a 250 mL volumetric flask, make up to the mark with distilled water and mix thoroughly — a dilution factor of 25.
2. Rinse and fill the burette with the standard $KMnO_4$; read the upper meniscus.
3. Pipette 25.0 mL of the diluted peroxide into a conical flask; add about 20 mL of dilute sulphuric acid.
4. Titrate in the cold with swirling, dropwise near the end, to the first **pale pink persisting 30 seconds**. Oxygen effervesces throughout — that is the analyte leaving as gas, exactly as the equation says.
5. Repeat until two titres agree within 0.1 mL; average the concordant titres.

4. Worked calculation (model data)

STRENGTH RESULT

Standard $KMnO_4$: 0.0200 M. Aliquot of diluted peroxide: 25.0 mL. Concordant titre: 22.5 mL.

Moles of $KMnO_4 = 0.0200 \times 0.0225 = 4.50 \times 10^{-4}$ mol.

Moles of $H_2O_2 = (5/2) \times 4.50 \times 10^{-4} = 1.125 \times 10^{-3}$ mol in the 25.0 mL aliquot (ratio 2 : 5).

Concentration of the diluted solution $= 1.125 \times 10^{-3} / 0.0250 = 0.0450$ M.

Concentration of the supplied solution $= 0.0450 \times 25$ (dilution factor) $= 1.125$ M.

Strength $= 1.125 \times 34.0 =$ **38.25 g L^{-1}** (about 3.8 % w/v).

> Volume strength: 1.125 mol H_2O_2 per litre releases 1.125/2 = 0.5625 mol O_2, and 0.5625 × 22.4 = 12.6 L of O_2 at STP per litre of solution — the peroxide is **12.6-volume**.

5. Observation table (as recorded)

TRIAL	INITIAL READING (ML)	FINAL READING (ML)	TITRE (ML)
1 (rough)	0.0	23.1	23.1
2	0.0	22.5	22.5
3	0.0	22.5	22.5

Check the two answers against each other before leaving the bench: 38.25 g L^{-1} ÷ 3.04 g L^{-1} per ‘volume’ ≈ 12.6 — the mass figure and the volume figure are the same fact in two dialects, and they must agree.

6. Precautions

- Never warm the flask: heat decomposes H_2O_2 (and dust or MnO_2 traces catalyse it), so peroxide would vanish without consuming permanganate — the strength would read low.
- Make the dilution carefully and mix the flask thoroughly; every later number is multiplied by the dilution factor, and so is every dilution error.
- Acidify before titrating: in neutral solution permanganate falls to MnO_2 and the 2 : 5 ratio no longer holds.
- Titrate the diluted solution reasonably promptly — dilute peroxide in a dusty laboratory loses strength measurably over a day.
- Rinse the burette out at once afterwards; permanganate left standing stains and blocks the jet with MnO_2.

7. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. In this titration, is hydrogen peroxide acting as an oxidising or a reducing agent? Justify from the equation.
2. Why is the peroxide diluted before titration instead of being titrated as supplied?
3. With the same dilution (10.0 mL to 250 mL), a 25.0 mL aliquot needs 20.0 mL of 0.0200 M $KMnO_4$. Find the strength in g L^{-1} and the volume strength.

4. Why is this flask kept cold when the oxalate flask of Experiment 7 had to be hot?
5. Show that a ‘10-volume’ peroxide is about 30 g L^{-1}.
6. The end-point pink of this titration sometimes fades within a minute. Give a reason, and state the reading rule that protects the result.

Answer key

1. A reducing agent: peroxide’s oxygen (−1 oxidation state) is oxidised to O_2 (0), while MnO_4^- is reduced to Mn^{2+}. Against weaker oxidants peroxide plays the opposite role; against permanganate it is the reductant.
2. Undiluted, the titre would exceed the burette’s capacity many times over and a single drop would be a negligible fraction of the total — poor precision and wasted reagent. A known 25-fold dilution brings the titre into the accurate middle of the burette.
3. Moles $KMnO_4$ = 0.0200 × 0.0200 = 4.00 × 10^{-4}; moles H_2O_2 = (5/2)(4.00 × 10^{-4}) = 1.00 × 10^{-3} in 25.0 mL → 0.0400 M diluted → × 25 = 1.00 M supplied → 1.00 × 34.0 = 34.0 g L^{-1}. Oxygen: 0.500 mol per litre → 0.500 × 22.4 = 11.2 L → 11.2-volume.
4. The oxalate reaction is intrinsically slow when cold, so heat supplied the rate. The peroxide reaction is already fast in the cold, and heating would decompose the analyte itself ($2\ H_2O_2 \rightarrow 2\ H_2O + O_2$) before the burette could count it.
5. 10 L of O_2 at STP = 10/22.4 = 0.446 mol O_2, which comes from 2 × 0.446 = 0.893 mol H_2O_2 = 0.893 × 34.0 ≈ 30 g — so 10-volume ≈ 30 g L^{-1} (≈ 3 % w/v).
6. Traces of reducing matter and dust keep consuming the slight excess of permanganate, and dilute peroxide solutions also decompose slowly — the pink drains. The rule, as in Experiment 8: read the first pink that persists for 30 seconds.

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